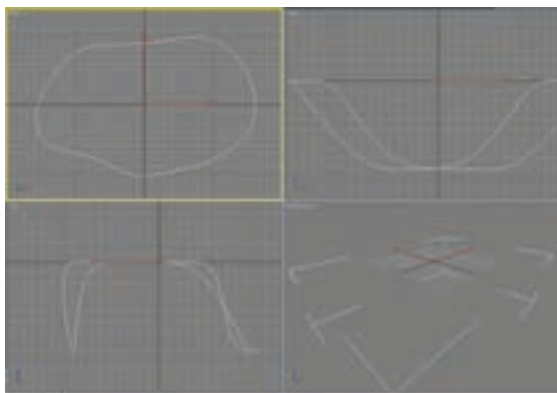


2.7 Using Splines

In 3ds Max, a spline is a Bezier-based line that defines a closed or open shape. Splines are typically used as building blocks to create other objects within 3ds Max, but they can also be rendered to create cylindrically shaped



Splines

objects. Splines can also be used as animation paths or to control inverse kinematic (IK) chains. Splines are created by using the Shapes tab of the Create panel. You can draw free-form lines or create a number of predefined shapes, such as circles, rectangles, helixes, and text.

Drawing Lines

The Line object under the Create panel will allow you to draw a line in a viewport. Lines are drawn by clicking in the viewport to lay down the vertices. Clicking the viewport creates the type of vertex defined under Initial Type, while clicking and dragging creates the type of vertex defined under Drag Type. This allows you to shape the line as it is drawn, reducing the amount of editing.

Editing Splines

Splines created by using the Line object have editing tools built in. Splines created by using any of the parametric tools such as Circle, Helix, or Text can be edited by applying an Edit Spline modifier to the stack. Splines have three classes of sub-objects: vertex, segment, and spline. Vertices are the points that define the line, segments are the portion of the spline between two vertices, and a spline is the entire line. A Spline object is not limited to a single line. A vertex on a spline can be one of four types a Corner, a Bezier corner, a Bezier and a Smooth. Besides manipulating the geometry itself, detail can be added to splines, and splines can be connected by using the tools located under the Geometry tab.

Attach: Attaches another spline object to the current spline

Weld: Welds two end points together to connect two splines as one

Insert: Inserts a new vertex by clicking on a spline